

Wazuh SIEM + EDR Deployment Guide

Purpose

Wazuh serves as the Security Information and Event Management (SIEM) platform and Endpoint Detection and Response (EDR) system for the Virtual SOC environment. It centralizes logs, correlates events, monitors endpoints, visualizes alerts, and provides dashboards for SOC analysis.

Wazuh is deployed on an Ubuntu 22.04 server and integrates with: - pfSense (system logs) - Suricata (EVE JSON alerts) - Kali, victim hosts, and other endpoints (Wazuh agents) - WireGuard operator connections

1. Wazuh Architecture Components

A full Wazuh installation includes: - **Wazuh Manager**
Correlates events, processes rules, generates alerts.

- **Wazuh Indexer**
Stores and indexes security data.
- **Wazuh Dashboard**
Web interface for SOC teams.

All three are automatically installed using the provided installation script.

2. Installation Steps

Step 1 — Download Installer

```
curl -sO https://packages.wazuh.com/4.x/wazuh-install.sh
```

Step 2 — Launch Installation

```
sudo bash wazuh-install.sh -a
```

This installs: - Manager - Indexer - Dashboard - Required dependencies

Step 3 — Access the Dashboard

Upon completion, the script outputs access credentials:

```
https://<WAZUH-IP>:443  
User: admin  
Password: <generated-password>
```

Confirmed functionality by logging in through browser.

3. System Requirements

Recommended

- Ubuntu Server 22.04 LTS
- 4–6 GB RAM minimum
- 2 vCPUs
- 40 GB disk

Networking

- Must be reachable by pfSense for syslog-ng forwarding (UDP 514)
 - Must be reachable by Agents (port 1514/1515)
 - Dashboard exposed on HTTPS 443
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4. Agent Deployment

Wazuh agents provide endpoint telemetry:

File integrity monitoring, log ingestion, malware detection, vulnerability detection, and system info.

Step 1 — Generate Agent Installer

Dashboard → **Agents** → **Deploy New Agent**

Step 2 — Configure

Provide: - Server IP (Wazuh Manager IP) - Agent name

Step 3 — Install on Endpoint

Copy the provided script into any Linux/Windows/macOS endpoint.

Step 4 — Confirm Agent Status

```
sudo systemctl status wazuh-agent
```

Manager side: - Dashboard → Agents → Agent List

Agent transitions to **Active** once connected and authenticated.

5. Log Ingestion Sources

Wazuh ingests:

1. Suricata Alerts

Forwarded from pfSense via syslog-ng as raw EVE JSON. These become: - Network attack alerts - Scan detections - Malware signatures - Recon activity

2. pfSense System Logs

Optional: forwarded by syslog-ng.

3. Endpoint Telemetry

From installed Wazuh agents on hosts. Includes: - Authentication logs - OSSEC/FIM logs - Vulnerability scans - Command/process monitoring

4. Custom Logs

Any log can be shipped with custom decoders.

6. Dashboard Features

The web interface provides: - Live alert feed - Event correlation view - MITRE ATT&CK mapping - File integrity dashboards - Vulnerability summary - Endpoint status monitoring - Network threat overview (Suricata events) - Search/Query bar using Wazuh Query Language

Example Query Resource:

<https://documentation.wazuh.com/current/user-manual/wazuh-dashboard/queries.html>

7. SIEM Pipeline Verification

Step 1 — Confirm Suricata Alerts Arrive

On Wazuh Manager:

```
sudo tcpdump -i eth0 -n -s 0 -A 'udp port 514 and host <pfsense-wan-ip>'
```

Step 2 — Generate Test Alert

From any host:

```
curl http://testmyids.com
```

Expected: - Alert appears in **Dashboard** → **Security Events** within seconds.

Step 3 — Agent Alert Verification

Stop a service on a monitored endpoint:

```
sudo systemctl stop ssh
```

Wazuh should produce: - Service status change alert - Possible security warning depending on ruleset

8. Integration with WireGuard

When analysts connect via WireGuard: - Dashboard accessible only through SIEM private IP - No exposure of Wazuh to the public Internet - Secure access from operator workstation

WireGuard configuration is detailed separately.

9. Operational Use in VSOC

Wazuh enables:

- Threat detection
- Incident response workflows
- Compliance checks

- Host-level telemetry
- Correlation across NIDS + EDR data

Combined with Suricata:

- Network + host correlation
- Full-picture attack chain monitoring

For instruction/demos:

- Runs Nmap scans
- Executes exploits against Metasploitable
- Observes real-time SIEM detections

10. Summary of Responsibilities

Wazuh serves as the central SOC platform and is responsible for: - Storing and indexing all security logs - Correlating Suricata and endpoint telemetry - Providing dashboards and alerting for analysis - Raising high-confidence detections - Supporting full lab demonstration workflows

The SIEM sits at the heart of the Virtual SOC ecosystem, turning raw data into actionable visibility for analysts.